7	(a)	State experimental evidence to suggest that the process of radioactive decay is		
		(i)	random;	
			…………………………………………………………………………………………	
			…………………………………………………………………………………………	[1]
			The **points on a count rate against time graph scatter about the (exponential) line of best fit.** **OR** **Measured count rate fluctuates** from instant to instant in time**.**	**1**
		(ii)	spontaneous.	
			…………………………………………………………………………………………	
			…………………………………………………………………………………………	[1]
			The rate of decrease of the measured count rate is unaffected by external stimuli and changes in physical conditions. **OR** **Repeated experiments under different physical conditions result in no change in the rate of decrease of the measured count rate.**	**1**
	(b)	Uranium-238 decays into lead-206 by several stages. Lead-206 is a stable isotope. The overall decay can be represented by the following equation: $^{238}_{92}\text{U} \rightarrow {}^{206}_{82}\text{Pb} + \text{decay products}$ It is suggested that **all** of the decay products are alpha particles. Use the equation to show that this cannot be correct.		
				[2]
		Proof by Contradiction: Suppose all the decay products are alpha particles and total number of alpha particles produced is x. $^{238}_{92}\text{U} \rightarrow {}^{206}_{82}\text{Pb} + x\,{}^{4}_{2}\text{He}$ **For nucleon number to be conserved**: $238 = 206 - 4x \Rightarrow x = 8$ **For proton number to be conserved:** $92 = 82 - 2x \Rightarrow x = 5$ Need 8 alpha particles to balance the total number of nucleons, but 5 alpha particles to balance the total number of protons in the equation. Hence, it is not possible for all		**1**

		decay products to be alpha particles. OR alpha-particle: ${}^{4}_{2}He$ $\Longrightarrow$ ratio of nucleon no. to proton no. = 2:1 Total number of nucleons in the decay products, A = 236 – 206 = 32 Total number of protons in the decay products, B = 92 – 82 = 10 ratio A:B = 32:10 ≠ 2:1 Therefore not all the decay products are alpha particles. OR alpha-particle: ${}^{4}_{2}He$ $\Longrightarrow$ 2 protons and 2 neutrons Total number of protons in the decay products = 92 – 82 = 10 Total number of neutrons in the decay products = (238-92) – (206-82) = 22 Since an alpha-particle has equal number of protons and neutrons, but the total number of protons and neutrons are different in the decay products, hence not all the decay products are alpha-particles.		**1**
	(c)	Technetium-99, ${}^{99}_{43}Tc$, decays to ruthenium-99, ${}^{99}_{44}Ru$. The half-life of technetium-99 is 4.00 x 10^{6} years. Ruthenium-99 is a stable nuclide.		
		(i)	Write down the nuclear equation representing this decay. State also the name(s) of the products other than ruthenium-99 that is/are formed.	
			Equation:	
			Name(s) of additional product(s):	[2]
			${}^{99}_{43}Tc \rightarrow {}^{99}_{44}Ru + {}^{0}_{-1}e$ **Beta-particle <u>AND</u> antineutrino/neutrino.**	**1** **1**
			•	
		(ii)	On the axes of Fig. 7.1, sketch a graph to show how the ratio $$R = \frac{\text{number of ruthenium-99 nuclei}}{\text{number of technetium-99 nuclei}}$$ will change in a sample with time *t*. Take *t* = 0 to be the instant of creation of ruthenium-99.	

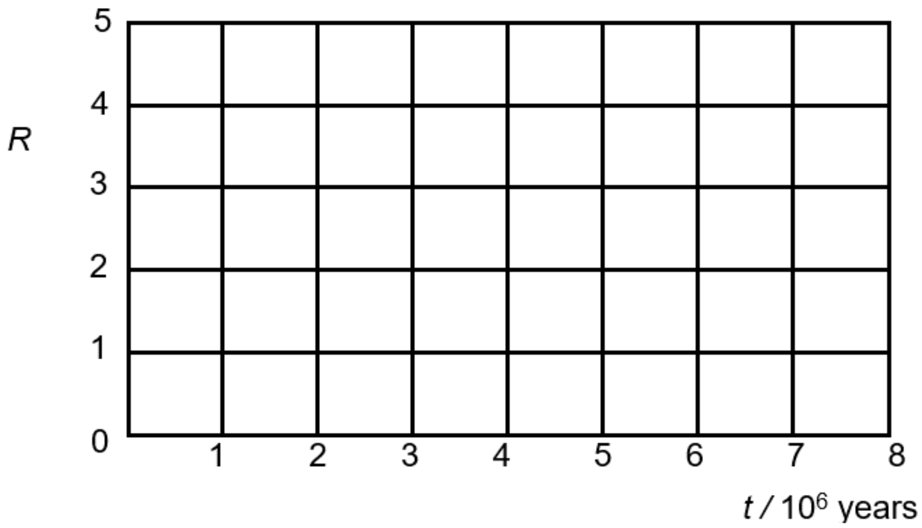
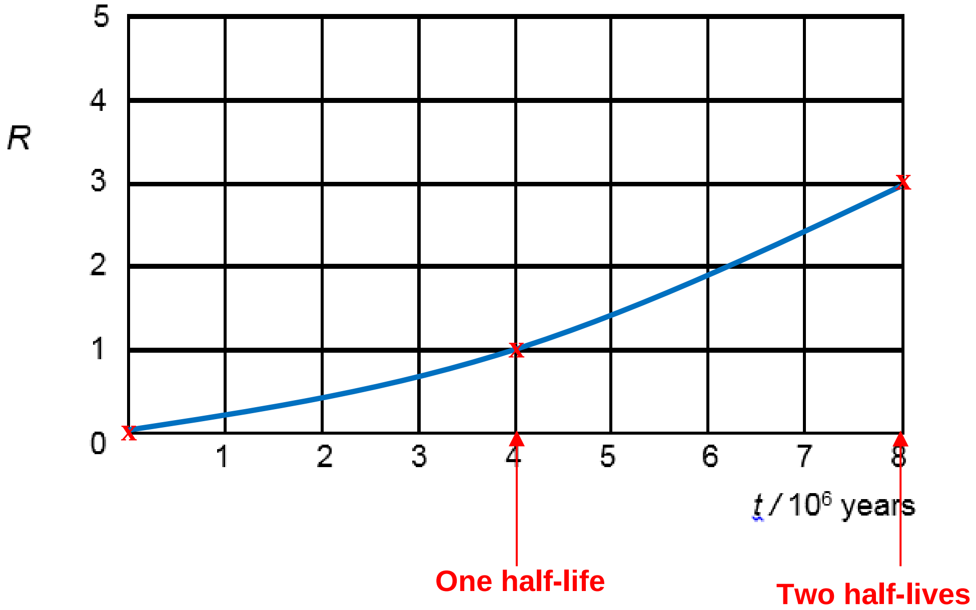


Fig. 7.1 [2]

Ru should be increasing, Tc decreasing with time.
Ru:Tc
0 : N at t=0
N/2 : N/2 after one half-life
3N/4 : N/4 after two half-lives
7N/8 : N/8 after three half-lives
etc…

Therefore,
R = 0 at t=0 → plot (0,0)
R = 1 at t=4x10^6 years → plot (4,1)
R = 3 at t=8x10^6 years → plot (8,3)

Since N is an exponential function with time, the ratio R will also be an exponential function. So graph of R is an exponential curve.



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Section B

Answer **one** question from this section.

8	**(a)**	Explain what is meant by an *electric field*.